

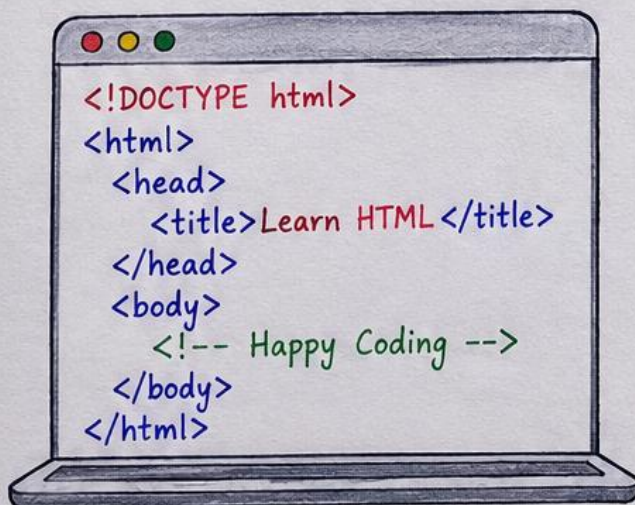
Complete Handwritten Notes

HTML

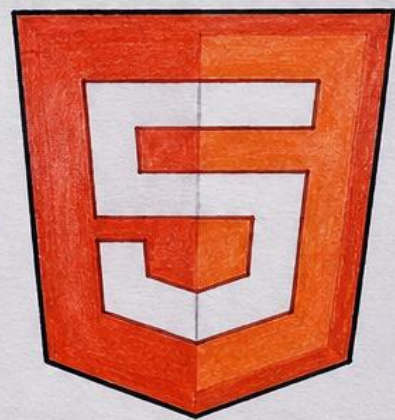
Zero to Hero

From Basics to Advanced

- Simple & Easy Notes
- Exam & Interview Focused
- Concepts with Examples
- Diagrams & Code Snippets
- Perfect for Beginners



HTML



"Build the Web
One Tag at a Time"

Code Learn Create Repeat

By :-

Narayan Prasad



Zero to Hero

From Basics to Advanced

1. What is HTML ?

- HTML (HyperText Markup Language) is the standard language used to create web pages.
- It defines the structure and content of a web page using tags and elements.
- HTML is not a programming language, it is a markup language.
- It works with CSS (for styling) and JavaScript (for interactivity) to create complete websites.
- It is simple, easy to learn and widely used in web development.

Key Points

- Stands for **HyperText Markup Language**
- Used to create web pages
- Defines structure and content
- Works with CSS and JavaScript
- Easy to learn
- Used in almost all websites

2. History of HTML (Short)

- HTML was created by Tim Berners-Lee in 1991 at CERN.
- First version was very simple with limited tags.
- Over time, many versions were released (HTML 2.0, 3.2, 4.01, XHTML, HTML5).
- HTML5 is the latest and most widely used version.

Major Versions

- HTML 1.0 → 1991
- HTML 2.0 → 1995
- HTML 3.2 → 1997
- HTML 4.01 → 1999
- XHTML → 2000
- HTML5 → 2014 (latest)

3. Features of HTML

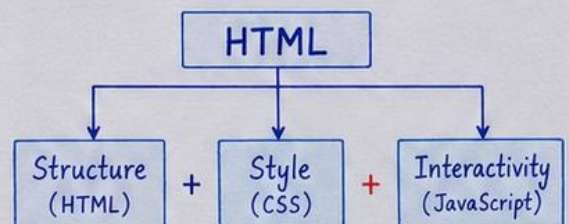
- Simple and easy to learn
- Platform independent
- Supports multimedia (images, audio, video)
- Works with CSS and JavaScript
- Widely supported by all browsers
- Used to create static and dynamic web pages

Common Web Browsers

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari
- Opera
- Brave
- and more...

4. Real World Example

- Every website you open (like Google, YouTube, Amazon) is built using HTML.
- HTML helps to organize the content like headings, paragraphs, images, links etc.



HTML is the foundation of every website

Zero to Hero

From Basics to Advanced

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3. First HTML Example

HTML Code :

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Web Page </title>
  </head>
  <body>
    <h1>Hello, World!</h1>
    <p>This is my first HTML page. </p>
  </body>
</html>
```

Output :

Hello, World!

This is my first HTML page.

4. How HTML Works ?

- A web browser reads the HTML code.
- It interprets the tags and displays the content on the screen.
- HTML provides the structure (skeleton) of the webpage.
- CSS is used for styling and JavaScript is used for interactivity.

HTML is the foundation of every website

1. HTML Document Structure

- Every HTML document follows a basic structure.
- It tells the browser the document type, language, metadata and visible content.
- The basic structure is same for almost all web pages.
- It is the starting point of any HTML page.

Key Tags

<code><!DOCTYPE html></code>	- Declares document type
<code><html></code>	- Root element
<code><head></code>	- Contains meta information
<code><title></code>	- Sets the title of the page
<code><body></code>	- Contains visible content

2. Example of HTML Structure

HTML Code :

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>My Web Page</title>
  </head>
  <body>
    <h1>Welcome to HTML</h1>
    <p>This is a simple HTML page.</p>
  </body>
</html>
```

Output :

Welcome to HTML

This is a simple HTML page.

3. Explanation of Each Part

<code><!DOCTYPE html></code>	- Tells the browser this is an HTML5 document.
<code><html></code>	- Root element that contains all other elements.
<code><head></code>	- Contains meta information (not visible on page).
<code><title></code>	- Title shown in the browser tab.
<code><body></code>	- Contains the actual content (visible on page).

4. Important Points

- HTML tags are not case sensitive, but it is good practice to write in lowercase.
- The `<title>` tag is important for SEO.
- You can add CSS (in `<head>`) and JavaScript (in `<head>` or `<body>`).
- Every HTML page must have one `<html>`, `<head>` and `<body>` tag.

HTML Tags & Elements

1. What are HTML Tags ?

- HTML tags are special keywords written inside angle brackets < >.
- They tell the browser what type of content to display.
- Most tags come in pairs (opening and closing).

Example :

```
<p>This is a paragraph.</p>
```

3. Types of HTML Tags

- Paired Tags** : Have opening and closing tag.
Example : `<div> ... </div>`
- Self-closing (Empty) Tags** : Do not have closing tag.
Example : `<br>`, `<hr>`, `<img>`
- Nested Tags** : A tag inside another tag.
Example : `<div><p>Text</p></div>`

5. Commonly Used HTML Tags

Tag	Purpose
<code><html></code>	Defines the root of HTML document
<code><head></code>	Contains meta information
<code><title></code>	Sets the title of the page
<code><body></code>	Contains visible content
<code><h1></code> to <code><h6></code>	Headings
<code><p></code>	Paragraph
<code><a></code>	Link
<code></code>	Image
<code><div></code>	Division (container)
<code></code>	Inline container
<code>
</code>	Line break
<code><hr></code>	Horizontal line

2. What are HTML Elements ?

- An HTML element is the complete unit consisting of an opening tag, content and a closing tag.
- Element = Opening tag + Content + Closing tag

Example :

```
<h1>Hello World</h1>
```

↑ Opening Tag ↑ Content ↑ Closing Tag

4. Tag vs Element

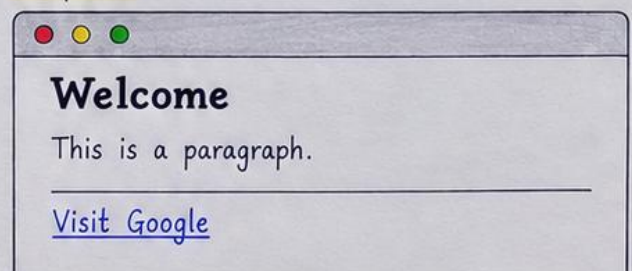
Tag	Element
Only the mark (like <code><p></code>)	Complete unit (like <code><p>Text</p></code>)
Just a keyword	Tag + content + tag
Example: <code><h1></code>	Example: <code><h1>Hello</h1></code>

6. Example: Using Multiple Tags

HTML Code :

```
<html>
  <head>
    <title>My Page </title>
  </head>
  <body>
    <h1>Welcome </h1>
    <p>This is a paragraph. </p>
    <hr>
    <a href="https://www.google.com">
      Visit Google
    </a>
  </body>
</html>
```

Output :



Headings, Paragraphs & Text Formatting

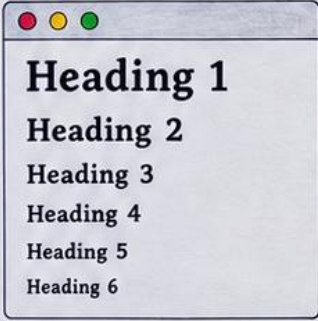
1. Headings in HTML

- HTML provides 6 levels of headings from `<h1>` to `<h6>`.
- `<h1>` is the largest and `<h6>` is the smallest.
- Headings are used to define important titles and subtitles in a webpage.

Example :

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

Output :



Heading 1
Heading 2
Heading 3
Heading 4
Heading 5
Heading 6

2. Paragraphs in HTML

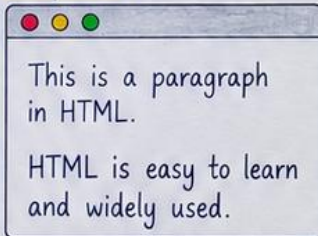
- The `<p>` tag is used to define a paragraph.
- Browsers automatically add some space before and after a paragraph.

Example :

```
<p>This is a paragraph
in HTML.</p>

<p>HTML is easy to learn
and widely used.</p>
```

Output :



This is a paragraph
in HTML.

HTML is easy to learn
and widely used.

3. Line Break and Horizontal Rule

- The `<br>` tag is used to insert a line break.
- The `<hr>` tag is used to add a horizontal line (thematic break).

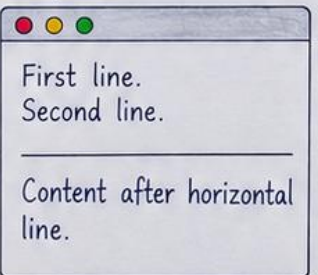
Example :

```
<p>First line.<br>
  Second line.</p>

<hr>

<p>Content after horizontal
line.</p>
```

Output :



First line.
Second line.

Content after horizontal
line.

4. Text Formatting Tags

- HTML provides several tags to format text.
- These tags help to make text bold, italic, underlined, highlighted, etc.

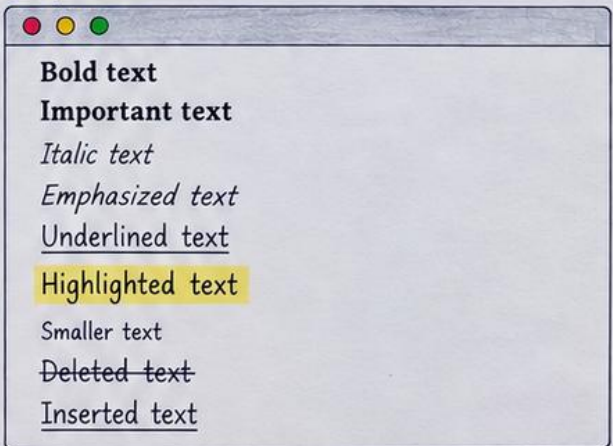
Tag	Purpose
<code></code>	Bold text (visual only)
<code></code>	Important text (semantic)
<code><i></code>	Italic text (visual only)
<code></code>	Emphasized text (semantic)
<code><u></code>	Underlined text
<code><mark></code>	Highlighted text
<code><small></code>	Smaller text
<code></code>	Deleted text (strikethrough)
<code><ins></code>	Inserted text (underlined)

5. Example of Text Formatting

HTML Code :

```
<p><b>Bold text</b></p>
<p><strong>Important text</strong></p>
<p><i>Italic text</i></p>
<p><em>Emphasized text</em></p>
<p><u>Underlined text</u></p>
<p><mark>Highlighted text</mark></p>
<p><small>Smaller text</small></p>
<p><del>Deleted text</del></p>
<p><ins>Inserted text</ins></p>
```

Output :



Bold text
Important text
Italic text
Emphasized text
Underlined text
Highlighted text
Smaller text
~~Deleted text~~
Inserted text

Links & Images

1. Links in HTML

- The `<a>` tag is used to create a hyperlink.
- Links are used to navigate to other web pages, files, or sections within a page.
- The `href` attribute specifies the destination URL.
- Links can be to external websites, internal pages or email addresses.

Syntax :

```
<a href="URL">Link Text</a>
```

3. Example of Links

HTML Code :

```
<h2>Links Example</h2>
<p><a href="https://www.google.com">
  Visit Google</a></p>
<p><a href="about.html">About Page</a></p>
<p><a href="#section1">Go to Section</a></p>
<p><a href="mailto:test@gmail.com">
  Send Email</a></p>
```

2. Types of Links

Type	Example
External Link	<code>Google</code>
Internal Link	<code>About</code>
Link to Section	<code>Go to Section</code>
Email Link	<code>Send Email</code>
Open in New Tab	<code>Google</code>

Output :

Links Example

[Visit Google](https://www.google.com)
[About Page](about.html)
[Go to Section](#section1)
[Send Email](mailto:test@gmail.com)

4. Images in HTML

- The `<img>` tag is used to insert an image in a web page.
- The `src` attribute specifies the path to the image.
- The `alt` attribute provides alternative text if the image cannot be displayed.
- Images can be in different formats like JPG, PNG, GIF, etc.

Syntax :

```

```

6. Important Points

- Use meaningful link text.
- Use `alt` attribute for all images.
- Use appropriate image formats (JPG, PNG, WebP).
- Use `target="_blank"` to open links in new tab.
- Keep image file size optimized for better performance.

5. Example of Image

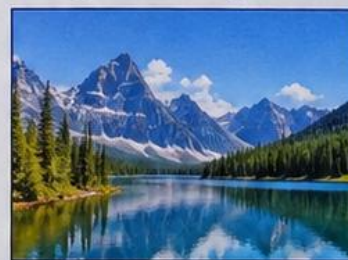
HTML Code :

```
<h2>Image Example</h2>

```

Output :

Image Example



1. What are Lists ?

- Lists are used to group related items in a structured way.
- HTML provides 3 types of lists :
 1. Ordered List (`<ol>`)
 2. Unordered List (`<ul>`)
 3. Description List (`<dl>`)
- List items are defined using the `<li>` tag (List Item).

3. Unordered List

- An unordered list displays items with bullets instead of numbers.
- It uses the `<ul>` tag and `<li>` for each item.
- You can change the bullet style using the type attribute (disc, circle, square).

Example :

```
<ul type="disc">
  <li>Apple</li>
  <li>Banana</li>
  <li>Orange</li>
</ul>
```

Output :

- 
- Apple
 - Banana
 - Orange

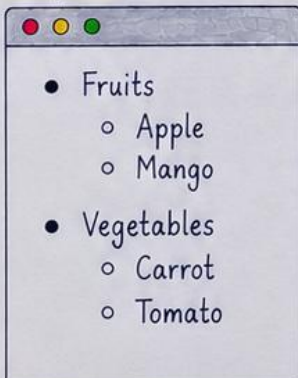
5. Nested Lists

- You can also create a list inside another list (nested list).
- It is useful for sub-categories.

Example :

```
<ul>
  <li>Fruits
    <ul>
      <li>Apple</li>
      <li>Mango</li>
    </ul>
  </li>
  <li>Vegetables
    <ul>
      <li>Carrot</li>
      <li>Tomato</li>
    </ul>
  </li>
</ul>
```

Output :

- 
- Fruits
 - Apple
 - Mango
 - Vegetables
 - Carrot
 - Tomato

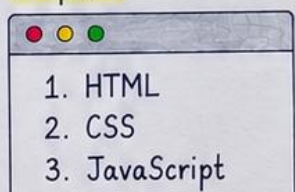
2. Ordered List

- An ordered list displays items in a numbered sequence.
- It uses the `<ol>` tag and `<li>` for each item.
- You can change the numbering type using the type attribute (1, A, a, I, i).

Example :

```
<ol type="1">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

Output :

- 
1. HTML
 2. CSS
 3. JavaScript

4. Description List

- A description list is used to display terms and their descriptions.
- It uses the `<dl>` tag, `<dt>` for the term and `<dd>` for the description.

Example :

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup
    Language</dd>
  <dt>CSS</dt>
  <dd>Used for styling
    web pages</dd>
</dl>
```

Output :



HTML
HyperText Markup
Language

CSS
Used for styling
web pages

6. Important Points

- Use `<ol>` for numbered lists.
- Use `<ul>` for bullet lists.
- Use `<dl>`, `<dt>` and `<dd>` for description lists.
- Use the type attribute to change numbering or bullet style.
- Lists can be nested.
- Lists are widely used in menus, navigation bars and structured content.

Lists make content more organized and easy to read.

Tables in HTML

1. What are Tables ?

- Tables are used to display data in rows and columns.
- They are useful for showing structured data like tabular information.
- A table is created using the `<table>` tag.

2. Table Related Tags

Tag	Purpose
<code><table></code>	Defines a table
<code><tr></code>	Defines a table row
<code><th></code>	Defines a header cell (bold & centered)
<code><td></code>	Defines a data cell
<code><caption></code>	Adds a title to the table
<code><thead></code>	Groups header content
<code><tbody></code>	Groups body content
<code><tfoot></code>	Groups footer content

3. Basic Table Example

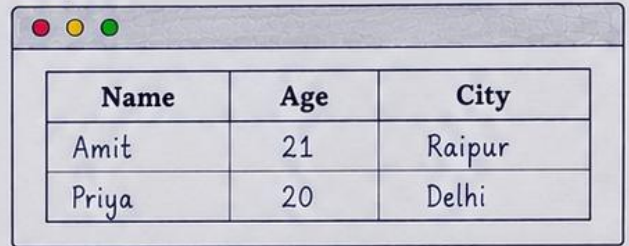
HTML Code :

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
  <tr>
    <td>Amit</td>
    <td>21</td>
    <td>Raipur</td>
  </tr>
  <tr>
    <td>Priya</td>
    <td>20</td>
    <td>Delhi</td>
  </tr>
</table>
```

5. Important Points

- Use `<th>` for header cells.
- Use `<td>` for data cells.
- Use `border`, `cellpadding`, `cellspacing` for styling (or CSS).
- Use `<caption>` to add a table title.
- Tables are mainly used for tabular data, not for layout.

Output :



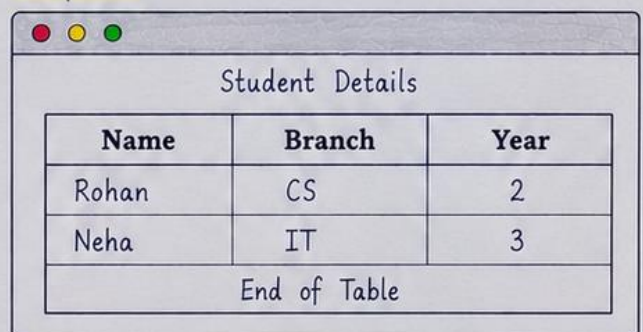
Name	Age	City
Amit	21	Raipur
Priya	20	Delhi

4. Table with Caption, thead, tbody, tfoot

HTML Code :

```
<table border="1">
  <caption>Student Details</caption>
  <thead>
    <tr>
      <th>Name</th>
      <th>Branch</th>
      <th>Year</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Rohan</td>
      <td>CS</td>
      <td>2</td>
    </tr>
    <tr>
      <td>Neha</td>
      <td>IT</td>
      <td>3</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td colspan="3">End of Table</td>
    </tr>
  </tfoot>
</table>
```

Output :



Student Details		
Name	Branch	Year
Rohan	CS	2
Neha	IT	3
End of Table		

Forms in HTML

1. What are Forms in HTML ?

- Forms are used to collect user input.
- They allow users to enter data and submit it to a server for processing.
- Forms are created using the `<form>` tag and contain different input elements like text fields, radio buttons, checkboxes, etc.
- Forms are widely used in login pages, registration pages, search boxes, etc.

2. The `<form>` Tag

- The `<form>` tag is a container for form elements.
- It has important attributes like `action` and `method`.
- The `action` attribute specifies the URL to which the form data will be sent.
- The `method` attribute specifies the HTTP method (GET or POST).

Syntax :

```
<form action="URL" method="GET|POST">
  <!-- form elements -->
</form>
```

3. Common Form Elements

Element	Purpose
<code><input></code>	Creates input fields
<code><label></code>	Defines a label for an input
<code><textarea></code>	Creates a multiline text input
<code><select></code>	Creates a drop-down list
<code><option></code>	Defines an option in a dropdown
<code><button></code>	Creates a clickable button
<code><fieldset></code>	Groups related form elements
<code><legend></code>	Defines a title for a fieldset

4. Example : Simple Registration Form

HTML Code :

```
<form action="/submit" method="post">
  <h2>Registration Form</h2>
  <label for="name">Name:</label>
  <input type="text" id="name" name="name">
  <br><br>
  <label for="email">Email:</label>
  <input type="email" id="email" name="email">
  <br><br>
  <label for="password">Password:</label>
  <input type="password" id="password" name="password">
  <br><br>
  <input type="submit" value="Register">
</form>
```

Output :

5. Important Points

- Use meaningful labels for form fields.
- The `name` attribute is important to identify form data on the server.
- Use the appropriate input type for each field.
- Use `required` attribute to make fields mandatory (if needed).
- Use proper form validation (HTML or JavaScript).
- Use POST method for sensitive data (e.g., login).
- Forms can also include checkboxes, radio buttons, select menus, file uploads, etc.

Form Input Types & Attributes

1. Common Input Types

- The `<input>` tag can have different types to accept different kinds of data.
- Each type provides a specific input control for the user.
- The type attribute defines the kind of input field.

Type	Description
text	Single line text input
password	Hides the entered text
email	Input for email address
number	Input for numeric values
tel	Input for telephone number
date	Date picker
radio	Select one option from a group
checkbox	Select multiple options
submit	Submit the form
reset	Reset the form fields
button	Clickable button (no default action)
file	Select a file from device
color	Color picker
range	Slider to select a value
search	Search input field
url	Input for URL

2. Examples of Input Types

HTML Code :

```
<form>
  <label>Name: <input type="text" name="name"></label><br>
  <label>Password: <input type="password" name="pass"></label><br>
  <label>Email: <input type="email" name="email"></label><br>
  <label>Age: <input type="number" name="age"></label><br>
  <label>Gender:
    <input type="radio" name="gender" value="M">Male
    <input type="radio" name="gender" value="F">Female
  </label><br>
  <label>Hobbies:
    <input type="checkbox" name="hobby" value="Reading">Reading
    <input type="checkbox" name="hobby" value="Music">Music
  </label><br>
  <label>Profile Picture: <input type="file" name="file"></label><br>
  <input type="submit" value="Submit">
  <input type="reset" value="Reset">
</form>
```

3. Output

4. Important Attributes

- Input fields can have additional attributes to control their behavior, validation and appearance.
- Some commonly used attributes are given below:

Attribute	Description
name	Specifies the name of the input field
id	Unique identifier for the input field
value	Specifies the default value
placeholder	Shows a hint inside the input field
required	Makes the field mandatory
readonly	Makes the field read-only
disabled	Disables the input field
min / max	Specifies minimum / maximum value (for number, date, range)
step	Specifies the allowed steps (for number, range)
size	Specifies the width (in characters)
maxlength	Specifies the maximum number of characters
multiple	Allows multiple values (for file, email, etc.)
pattern	Specifies a regular expression for validation
autofocus	Automatically focuses on this field when page loads

5. Example with Attributes

HTML Code :

```
<input type="text" id="username" name="username"
  placeholder="Enter username" required maxlength="20">
```

6. Important Points

- Use appropriate input types for better user experience.
- Use required and pattern for form validation.
- Use placeholder to give hints to users.
- Use proper labels with input fields for accessibility.
- Combine input types and attributes based on your needs.

Semantic HTML

1. What is Semantic HTML ?

- Semantic HTML uses meaningful tags that describe the purpose or meaning of the content.
- It makes the structure of a web page clear for both browsers and developers.
- It improves accessibility, SEO and code readability.
- It tells search engines what the content is about.
- **Example :** <header>, <nav>, <main>, <section>, <article>, <aside>, <footer>.

2. Semantic vs Non-Semantic Tags

Semantic Tag	Non-Semantic Tag
<header>	<div>
<nav>	<div>
<main>	<div>
<section>	<div>
<article>	<div>
<aside>	<div>
<footer>	<div>

3. Common Semantic Tags

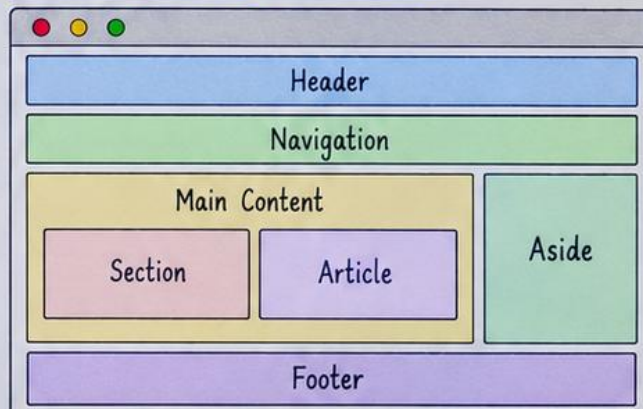
Tag	Purpose
<header>	Defines the header of a page or section
<nav>	Defines navigation links
<main>	Defines the main content of the page
<section>	Defines a section of content
<article>	Defines independent, self-contained content
<aside>	Defines sidebar or related content
<footer>	Defines the footer of a page or section
<figure>	Groups media content (like images)
<figcaption>	Defines a caption for <figure>
<details>	Defines additional details (collapsible)
<summary>	Defines a visible heading for <details>
<mark>	Highlights text
<time>	Defines a date or time
<progress>	Shows the progress of a task
<meter>	Represents a scalar measurement

4. Example : Semantic HTML Page

HTML Code :

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Semantic HTML Example</title>
  </head>
  <body>
    <header>
      <h1>My Website</h1>
    </header>
    <nav>
      <a href="#">Home</a> |
      <a href="#">About</a> |
      <a href="#">Contact</a>
    </nav>
    <main>
      <section>
        <h2>Welcome</h2>
        <p>This is a semantic HTML page.</p>
      </section>
      <article>
        <h2>Latest News</h2>
        <p>HTML5 introduces semantic tags.</p>
      </article>
      <aside>
        <p>Related Links</p>
      </aside>
    </main>
    <footer>
      <p>© 2025 My Website</p>
    </footer>
  </body>
</html>
```

5. Output (Layout View)



Audio, Video and Embedded Content

1. Adding Audio in HTML

- The `<audio>` tag is used to embed sound content in a webpage.
- It supports multiple audio formats like MP3, OGG and WAV.
- Use the controls attribute to show play, pause, volume, etc.
- You can also use autoplay, loop and muted attributes (when needed).

Syntax :

```
<audio controls>
  <source src="song.mp3" type="audio/mpeg">
  <source src="song.ogg" type="audio/ogg">
  Your browser does not support the audio tag.
</audio>
```

Output :



3. Important Attributes

Attribute	Description
controls	Displays playback controls
autoplay	Automatically starts playing
loop	Repeats the media
muted	Mutes the audio/video
width / height	Sets the size (mainly for video)
poster	Specifies thumbnail image for video
src	Specifies the media file path
type	Specifies the media file type

4. Supported File Formats

Media Type	Common Formats
Audio	MP3, OGG, WAV
Video	MP4, WebM, OGG
Browser Support	MP4 (widely supported)

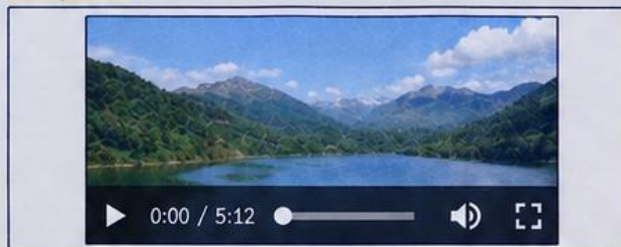
2. Adding Video in HTML

- The `<video>` tag is used to embed video content in a webpage.
- It supports formats like MP4, WebM and OGG.
- Use the controls attribute to show video controls.
- You can set width, height, autoplay, loop, poster (thumbnail), and muted attributes.

Syntax :

```
<video width="400" controls>
  <source src="video.mp4" type="video/mp4">
  <source src="video.webm" type="video/webm">
  Your browser does not support the video tag.
</video>
```

Output :



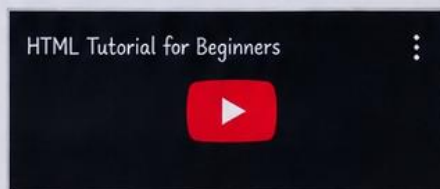
5. Embedding External Content

- The `<iframe>` tag is used to embed external content like YouTube videos, maps, or other websites.
- It specifies the source using the src attribute.
- You can set width, height and other attributes.

Example : YouTube Video

```
<iframe width="560" height="315"
  src="https://www.youtube.com/embed/dQw4w9WgXcQ"
  title="YouTube video" frameborder="0"
  allowfullscreen></iframe>
```

Output :



6. Important Points

- Use appropriate file formats for better compatibility.
- Always provide fallback content inside `<audio>` and `<video>`.
- Embedded content can come from other websites (e.g., YouTube, Google Maps).
- Large media files can slow down page loading, so optimize them.
- Use responsive dimensions for better user experience.

HTML Entities & Symbols

1. What are HTML Entities ?

- HTML entities are special codes used to display characters that are reserved in HTML or not available on the keyboard.
- They start with an ampersand (&) and end with a semicolon (;).
- Entities can be named (e.g., <) or numeric (e.g., <).
- They are used to display special symbols, spaces, and reserved characters correctly in the browser.

2. Why use HTML Entities ?

- To display reserved characters like <, >, & without being interpreted as HTML code.
- To display special symbols (©, ®, ™, etc.).
- To add spaces and other invisible characters.
- To ensure proper rendering across all browsers.

3. Common HTML Entities

Character	Entity Name	Numeric Code	Result
<	<	<	<
>	>	>	>
&	&	&	&
"	"	"	"
'	'	'	'
(space)	 	 	
©	©	©	©
®	®	®	®
™	™	™	™

4. Whitespace Entities

Entity	Description	Result
 	Non-breaking space	
 	En space (2 spaces wide)	
 	Em space (4 spaces wide)	
 	Thin space	
‌	Zero width non-joiner	
‍	Zero width joiner	

5. Common Symbols and Their Entities

Symbol	Entity Name	Numeric Code	Description
€	€	€	Euro sign
£	£	£	Pound sign
¥	¥	¥	Yen sign
₹	₹	₹	Rupee sign
×	×	×	Multiplication
÷	÷	÷	Division
±	±	±	Plus or minus
≤	≤	≤	Less than or equal
≥	≥	≥	Greater than or equal
≠	≠	≠	Not equal
≈	≈	≈	Approximately equal
∞	∞	∞	Infinity
√	√	√	Square root
α	α	α	Alpha (Greek)
β	β	β	Beta (Greek)
γ	γ	γ	Gamma (Greek)
δ	δ	δ	Delta (Greek)
π	π	π	Pi (Greek)
θ	θ	θ	Theta (Greek)

6. Example Usage

HTML Code :

```
<p>5 &lt; 10 and 10 &gt; 5</p>
<p>Copyright &copy; 2025</p>
<p>Price: &#8377; 100</p>
<p>Math: a &sup2; + b &sup2; = c &sup2;</p>
<p>Use &nbsp; for extra space.</p>
```

Output :

5 < 10 and 10 > 5
 Copyright © 2025
 Price: ₹ 100
 Math: $a^2 + b^2 = c^2$
 Use for extra space.

7. Important Points

- Use entities for reserved characters and special symbols.
- Both named entities (&name;) and numeric entities (&#code;) work.
- Entities are case-sensitive (e.g., © not ©).
- Use for non-breaking space (useful in layouts).
- Not all symbols are supported in older browsers.

HTML Attributes & Global Attributes

1. What are HTML Attributes ?

- Attributes provide additional information about HTML elements.
- They are always written inside the opening tag.
- Attributes usually come in name = "value" pairs.
- Multiple attributes can be used in a single tag, separated by spaces.

Syntax :

```
<tag attribute="value"> Content </tag>
```

2. Commonly Used Attributes

Attribute	Description
id	Specifies a unique id for an element
class	Specifies one or more class names
style	Specifies inline CSS styles
title	Provides extra information (tooltip)
href	Specifies the URL for a link
src	Specifies the source of an image, video, etc.
alt	Alternative text for an image
type	Specifies the type (e.g., input, button)
value	Specifies the value for input elements
name	Specifies the name of a form element
placeholder	Hint text inside an input field
required	Makes the field mandatory
disabled	Disables the element
readonly	Makes the field read-only
width	Specifies the width (e.g., image, video)
height	Specifies the height (e.g., image, video)
target	Specifies where to open a linked document

3. Example Usage

HTML Code :

```
<a href="https://www.google.com" target="_blank"
  title="Open Google">Visit Google</a>

<input type="text" name="username" placeholder="Enter name"
  required>
<button class="btn" style="color: blue;">Click Me</button>
```

4. Global Attributes

- Global attributes can be used on almost all HTML elements.
- They help in providing common functionality across elements.
- Some important global attributes are given below:

Attribute	Description
id	Unique identifier for the element
class	Specifies one or more class names
style	Inline CSS styles
title	Extra information (tooltip)
lang	Specifies the language of the element
dir	Specifies text direction (ltr or rtl)
hidden	Hides the element
tabindex	Specifies the tab order
accesskey	Keyboard shortcut to access the element
contenteditable	Makes the element editable
draggable	Specifies if the element is draggable
spellcheck	Enables spell checking
translate	Specifies whether the element should be translated

5. Example with Global Attributes

HTML Code :

```
<p id="info" class="text" title="This is a paragraph"
  lang="en" contenteditable="true" draggable="true">
  You can edit this text.
</p>
```

6. Important Points

- Attributes are case-insensitive (e.g., id or ID).
- Use meaningful id and class names.
- Global attributes can be used with most HTML elements.
- Use required, placeholder, and title for better user experience.
- Avoid using inline style for large projects (use CSS).
- Some attributes are specific to certain elements (e.g., href for <a>, src for).
- Using proper attributes improves accessibility and SEO.

HTML Block & Inline Elements

1. What are Block Elements ?

- Block elements always start on a new line.
- They take up the full width available (by default).
- They can contain other block or inline elements.
- Commonly used for layout and structure.

Example :

```
<div>
  <h2>This is a block element</h2>
  <p>It takes full width.</p>
</div>
```

2. Common Block Elements

Tag	Description
<div>	Defines a division or container
<p>	Defines a paragraph
<h1> - <h6>	Defines headings
<section>	Defines a section in a document
<article>	Defines independent content
<header>	Defines a header for a section/page
<footer>	Defines a footer
<nav>	Defines navigation links
	Defines an unordered list
	Defines an ordered list
	Defines a list item
<form>	Defines a form
<table>	Defines a table
<hr>	Defines a thematic break
<blockquote>	Defines a block quotation
<main>	Defines the main content

3. What are Inline Elements ?

- Inline elements do not start on a new line.
- They take only the required width.
- They cannot contain block elements (but can contain other inline elements).
- Commonly used for small pieces of content within a line.

Example :

```
<p>This is an <span style="color: blue;">inline element</span>
inside a paragraph.</p>
```

4. Common Inline Elements

Tag	Description
	Defines a span (inline container)
<a>	Defines a hyperlink
	Defines important text
	Defines emphasized text
	Defines bold text
<i>	Defines italic text
<u>	Defines underlined text
<small>	Defines smaller text
	Defines an image
<input>	Defines an input field
<label>	Defines a label
<button>	Defines a clickable button
<select>	Defines a drop-down list
<textarea>	Defines a multiline text input
<code>	Defines a piece of code

5. Block vs Inline Elements

Feature	Block Elements	Inline Elements
Line break	Start on a new line	Do not start on a new line
Width	Take full width (by default)	Take only required width
Can contain	Block or inline elements	Only inline elements
Common use	Layout and structure	Small pieces of content
Examples	<div>, <p>, <section>	, <a>,

6. Important Points

- Block elements help in structuring the webpage.
- Inline elements are used inside text or other elements.
- You can change the display behavior using CSS (e.g., display: block; or display: inline;).
- Not all elements are strictly block or inline (some are inline-block by default, e.g.,).
- Choose the right element based on the purpose and semantics.

HTML Best Practices + SEO Basics

1. HTML Best Practices

- Use semantic HTML elements (e.g., `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<footer>`).
- Maintain proper indentation and readability.
- Use meaningful tag and attribute names.
- Separate structure (HTML), style (CSS), and behavior (JavaScript).
- Always include the doctype declaration.
- Use lowercase tag names (recommended).
- Close all tags properly.
- Use alt attribute for images.
- Use comments to explain complex sections.

2. Semantic HTML vs Non-Semantic HTML

Semantic HTML	Non-Semantic HTML
Meaningful tags	No specific meaning
Improves accessibility	Does not improve accessibility
Better for SEO	Not SEO friendly
Examples: <code><header></code> , <code><nav></code> , <code><main></code> , <code><article></code>	Examples: <code><div></code> , <code></code>

3. Accessible HTML

- Use semantic elements.
- Provide alt text for images.
- Use proper heading structure (`h1` → `h6`).
- Ensure good color contrast.
- Use labels for form inputs.
- Make sure the website is keyboard navigable.

4. Clean and Maintainable Code

- Use consistent naming conventions.
- Avoid inline styles (use external CSS).
- Keep the code modular.
- Remove unnecessary code and comments.
- Use meaningful class and id names.
- Validate your HTML code using tools (e.g., W3C Validator).

5. SEO Basics

- SEO (Search Engine Optimization) helps your website rank higher in search engine results.
- Well-structured HTML improves SEO.
- Use semantic tags and proper headings.
- Write descriptive and unique title and meta tags.
- Use alt text for images.
- Create clean and readable URLs.
- Use internal links and proper anchor text.
- Make your website mobile-friendly.
- Improve page load speed (optimize images, avoid large files).

6. Important SEO Tags

Tag	Example	Purpose
<code><title></code>	<code><title>My Website</title></code>	Title shown in search results
<code><meta name="description"></code>	<code><meta name="description" content="Learn HTML"></code>	Short description of the page
<code><meta name="keywords"></code>	<code><meta name="keywords" content="HTML, Web"></code>	Keywords (less important now)
<code><link rel="canonical"></code>	<code><link rel="canonical" href="https://example.com"></code>	Preferred URL for the page
<code><meta name="viewport"></code>	<code><meta name="viewport" content="width=device-width, initial-scale=1.0"></code>	Makes page mobile-friendly

7. Important Points to Remember

- Use semantic HTML for better structure, accessibility and SEO.
- Always include a proper `<title>` and `<meta description>`.
- Use meaningful alt text for all images.
- Keep the code clean, consistent and well-indented.
- Avoid using too many non-semantic `<div>` elements.
- Make your website responsive and mobile-friendly.
- Validate your HTML code.
- Focus on content quality and proper structure.
- Follow best practices for a better user experience.

HTML Interview Questions & Important Points

1. What is HTML ?

Ans :

HTML (HyperText Markup Language) is the standard language used to create and structure web pages. It uses tags to define elements like headings, paragraphs, links, images, etc.

2. Difference between HTML and HTML5 ?

Feature	HTML	HTML5
Version	Older version	Latest version
Doctype	Complex	Simple <!DOCTYPE html>
Multimedia	Not supported (natively)	Supports <audio>, <video>
Graphics	Uses external tools (e.g., Flash)	Supports <canvas>, <svg>
Semantic Tags	Limited	Many new tags (e.g., <header>, <nav>)

3. What are Semantic Tags ?

Ans :

Semantic tags clearly describe the meaning of the content (e.g., <header>, <nav>, <section>, <article>). They improve readability, accessibility and SEO.

4. Difference between <div> and ?

Ans :

Feature	<div>	
Type	Block element	Inline element
Width	Takes full width	Takes only required width
Use	Used for layout and grouping	Used for small pieces of content
Example	...	 ...

5. What is the use of alt attribute in ?

Ans :

The alt attribute provides alternative text for an image if the image cannot be displayed. It also helps in accessibility and SEO.

6. What is the difference between id and class ?

Ans :

- id is unique and can be used only once per page.
- class can be used multiple times.
- id is mainly used for specific styling or JavaScript.
- class is used for grouping elements.

7. What are void elements ? Give examples.

Ans :

Void elements do not have a closing tag and do not contain any content.

Examples :
, <hr>, , <input>, <meta>, <link>

8. What is the difference between relative and absolute URL ?

Ans :

- Relative URL : Refers to a file within the same website (e.g., images/pic.jpg).
- Absolute URL : Specifies the full path including domain (e.g., https://www.example.com/images/pic.jpg).

9. How do you add comments in HTML ?

Ans :

Comments are used to add notes in the code and are not displayed in the browser.

Syntax : <!-- This is a comment -->

10. How can you make a webpage responsive ?

Ans :

- Use the viewport meta tag.
- Use relative units (% , em , rem , vw , vh).
- Use CSS media queries.
- Use flexible layouts (Flexbox or Grid).
- Optimize images and avoid fixed widths.

11. What is the purpose of the <meta> tag ?

Ans :

The <meta> tag provides metadata about the HTML document, such as character set, description, keywords and viewport settings.

Example : <meta charset="UTF-8">

HTML Cheat Sheet / Final Revision

1. Basic Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
  <title>Page Title</title>
</head>
<body>
  <!-- Content goes here -->
</body>
</html>
```

2. Commonly Used Tags

Tag	Purpose
<h1> - <h6>	Headings
<p>	Paragraph
<a>	Hyperlink
	Image
<div>	Division (block)
	Inline container
, ,	Lists
<table>	Table
<form>	Form
<input>	Input field
<button>	Button
<header>, <footer>	Page structure
<section>, <article>	Sectioning

3. Useful Meta Tags

```
<meta charset="UTF-8"> <!-- Character set -->
<meta name="viewport" content="width=device-width,
  initial-scale=1.0"> <!-- Responsive -->
<meta name="description" content="Page description">
<meta name="keywords" content="HTML, Web, Tutorial">
<meta name="author" content="Your Name">
```

4. Common Attributes

Attribute	Use
id	Unique identifier
class	Apply CSS / group elements
style	Inline CSS
src	Source of image, video, etc.
href	URL for a link
alt	Alternative text for images
title	Tooltip text
type	Type of input (text, button, etc.)
name	Name of form element
value	Initial value
target	Where to open the link (_blank, _self)
lang	Language of the document

5. HTML Entities (Common)

Entity	Code	Description
<	<	Less than
>	>	Greater than
&	&	Ampersand
"	"	Double quote
'	'	Single quote
©	©	Copyright
®	®	Registered trademark
™	™	Trademark
 	 	Non-breaking space

6. Doctype & HTML Version

- <!DOCTYPE html> is used for HTML5.
- It helps the browser render the page correctly.
- Always include it at the beginning.

7. Quick Tips

- Use semantic tags.
- Use meaningful id and class names.
- Keep code clean and indented.
- Use alt text for images.
- Make your webpage responsive (use meta viewport).
- Validate your HTML (e.g., W3C Validator).
- Follow best practices for accessibility and SEO.

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